

2024, Mar. 2024 Release LOC:Acute Adult

Infection: Musculoskeletal

Utilization Benchmarks: Multiple options available, see table below for options

Overview

Select Day

Initial review ⁽¹⁾

Episode Day 1

Episode Day 2

Episode Day 3-8

Episode Day 9

Discharge Screens ⁽³⁶⁾

Utilization Benchmarks

Notes

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Overview

Instruction:

This subset is for patients with osteomyelitis and septic arthritis. For patients with actual or suspected sepsis, refer to Infection: Sepsis subset.

Osteomyelitis: Osteomyelitis, an infection of the bone, may develop following an open fracture, a puncture wound, diabetic foot infection, surgical treatment of a closed injury, or hematogenous seeding bacteremia. Presenting symptoms of osteomyelitis are fever, pain, or signs of local inflammation. Patients with osteomyelitis require hospitalization for wound stabilization, if indicated, and initiation of intravenous anti-infectives.

Diagnosis of osteomyelitis is made by bone culture or imaging studies such as x-ray, computed tomography, or magnetic resonance imaging. Treatment includes intravenous anti-infective therapy. Thorough wound debridement, dead space management, wound protection, and adequate drainage are provided as indicated. Supplemental treatment also includes maximizing the nutritional, medical status, and vascular supply of the patient. The duration of anti-infective therapy is 4 to 6 weeks and is often completed in the post acute setting (Rao et al., Plast Reconstr Surg 2011, 127 Suppl 1: 177S-87S; Concia et al., Nucl Med Commun 2006, 27: 645-60; Calhoun and Manring, Infect Dis Clin North Am 2005, 19: 765-86).

Septic arthritis: Septic arthritis is an infection of the joint most commonly caused by staphylococci or streptococci. Risk factors for developing joint sepsis include rheumatoid or osteoarthritis, prosthetic joints, intravenous drug use, alcoholism, diabetes, previous intra-articular corticosteroid injection, or cutaneous ulcers. Native joint septic arthritis usually presents with a history of fever, warm, swollen, or tender joints, and decreased range of motion. Physical findings are usually minimal in an infection of the prosthetic joint, and swelling is usually slight. The most distinctive finding is a drainage presumed to originate in the underlying infected prosthetic joint. Patients with septic arthritis often require hospitalization for intravenous anti-infective therapy and drainage.

Evaluation of septic arthritis includes a thorough history and physical examination, imaging of the affected joint, and synovial fluid culture. The mainstay of treatment for these patients is anti-infective therapy and joint drainage. Initial anti-infective therapy should be broad-spectrum until the offending pathogen has been identified. Duration of parenteral therapy is approximately 15-21 days. Patients can then be transitioned to oral anti-infectives for the remainder of the 4 week treatment regimen. Osteomyelitis is a feared complication, especially in those with cartilaginous joint (e.g., sternoclavicular or sacroiliac) involvement. In these patients, the treatment regimen should continue for 6 weeks (Mathews et al., Lancet 2010, 375: 846-55; Garcia-De La Torre and Nava-Zavala, Rheum Dis Clin

North Am 2009, 35: 63-73; Mathews et al., Ann Rheum Dis 2007, 66: 440-5; Coakley et al., Rheumatology (Oxford) 2006, 45: 1039-41).

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The majority of the content in this subset is based on a limited number of key sources for osteomyelitis (Rao et al., Plast Reconstr Surg 2011, 127 Suppl 1: 177S-87S; Calhoun and Manning, Infect Dis Clin North Am 2005, 19: 765-86) and septic arthritis (Mathews et al., Lancet 2010, 375: 846-55; Garcia-De La Torre and Nava-Zavala, Rheum Dis Clin North Am 2009, 35: 63-73).

(Excludes PO medications unless noted)

Select Day, One:

● **Initial review, One:** ⁽¹⁾

(From arrival in ED through decision to admit)

(Symptom or finding within 24h)

● **ACUTE, ≥ One:** ⁽²⁾

– Osteomyelitis, actual or suspected, by bone scan, x-ray, CT or MRI ⁽³⁾

● **Septic arthritis and, One:** ^(3, 4)

● **Synovial fluid and, One:** ⁽⁵⁾

– WBC > 25,000/cu.mm(25x10⁹/L)-50,000/cu.mm(50x10⁹/L) and polys > 75%(0.75)

– WBC > 50,000/cu.mm(50x10⁹/L)

● **Episode Day 1, One:**

(Symptom or finding within 24h)

● **ACUTE, ≥ One:** ⁽²⁾

● **Osteomyelitis, actual or suspected, by bone scan, x-ray, CT or MRI and, ≥ One:** ⁽³⁾

– Anti-infective

– Bone culture scheduled or performed within 24h

● **Septic arthritis and, Both:** ^(3, 4, 6)

● **Synovial fluid and, One:** ⁽⁵⁾

– WBC > 25,000/cu.mm(25x10⁹/L)-50,000/cu.mm(50x10⁹/L) and polys > 75%(0.75)

– WBC > 50,000/cu.mm(50x10⁹/L)

● **Intervention, One:**

● **Native joint and, All:**

– Synovial fluid culture

– Anti-infective ⁽⁷⁾

– Joint drainage ⁽⁸⁾

– DVT prophylaxis, or patient ambulatory ⁽⁹⁾

● **Prosthetic joint and, All:**

– Synovial fluid culture

– Anti-infective ⁽⁷⁾

– Surgical evaluation, scheduled or performed within 24h ⁽¹⁰⁾

– DVT prophylaxis, or patient ambulatory ⁽⁹⁾

● **Episode Day 2, One:**

● **ACUTE, Both:**

● **Infection unresolved and, ≥ One:**

– Osteomyelitis ≤ 8d since admission

– Septic arthritis ≤ 7d since admission

– Anti-infective

● **Episode Day 3-8, One:**

● **ACUTE, One:**

● **Responder, discharge expected today if clinically stable last 12h, All:** ⁽¹¹⁾

– Afebrile

– Erythrocyte sedimentation rate (ESR) or C-reactive protein (CRP) levels decreasing ⁽³⁾

● **Infection resolving and, ≥ One:** ⁽¹²⁾

– Osteomyelitis and pain controlled or manageable ⁽¹³⁾

● **Septic arthritis and, Both:**

– Joint effusion resolving

– Pain controlled or manageable ⁽¹³⁾

● **Anti-infective, One:**

– Arranged

– Completed

– Not indicated

– Wound regimen established and adequate healing

- **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽¹⁴⁾
- **Complication or comorbidity and clinically stable for discharge, ≥ One:**
 - Adverse drug reaction resolving
 - Blood glucose within acceptable limits ⁽¹⁵⁾
 - *C. difficile* infection, symptoms resolving, and tolerating PO ⁽¹⁶⁾
 - Tolerating PO ⁽¹⁷⁾
- **Partial responder, not clinically stable for discharge and requires continued stay, Both:**
 - **Infection unresolved and, ≥ One:** ⁽³⁾
 - Osteomyelitis ≤ 8d since admission
 - Septic arthritis ≤ 7d since admission
 - **Anti-infective and, ≥ One:**
 - Temperature ≥ 100.4°F(38.0°C) ⁽¹⁸⁾
 - Bone culture pending ⁽¹⁹⁾
 - **Complex wound care, ≥ One:** ⁽²⁰⁾
 - **Wound care ≥ 3x/24h, ≤ 5d and, ≥ One:**
 - Requiring parenteral analgesic
 - ≥ 15 min in duration
 - Wound management teaching ≤ 24h ⁽²¹⁾
 - Hyperbaric oxygen and refractory osteomyelitis ≤ 3d ^(22, 23)
 - **Joint effusion and, Both:**
 - Increased joint effusion
 - Drainage performed within last 24h
 - Pain uncontrolled requiring change in medication, dose, or frequency ^(24, 25, 26)
 - Surgical debridement performed within last 24h
 - Adverse drug reaction requiring change in medication ≤ 24h ⁽²⁷⁾
 - **Blood glucose ≤ 24h, All:** ⁽²⁸⁾
 - **Finding, ≥ One:**
 - > 250 mg/dL(13.9 mmol/L)
 - < 70 mg/dL(3.9 mmol/L)
 - Blood glucose monitoring
 - Insulin adjustment or dose held ^(28, 29)
 - ***C. difficile* infection, actual or suspected and, Both:** ⁽³⁰⁾
 - Anti-infective (includes PO) ≤ 5d ^(31, 32)
 - Inadequate oral intake ⁽³³⁾
 - Inadequate oral intake and IV fluid ^(34, 35)
- **Episode Day 9, One:**
 - **ACUTE, One:**
 - **Responder, discharge expected today if clinically stable last 12h, All:** ⁽¹¹⁾
 - Afebrile
 - Erythrocyte sedimentation rate (ESR) or C-reactive protein (CRP) levels decreasing ⁽³⁾
 - Osteomyelitis resolving and pain controlled or manageable ^(12, 13)
 - **Anti-infective, One:**
 - Arranged
 - Completed
 - Not indicated
 - Wound regimen established and adequate healing
 - **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽¹⁴⁾
 - **Complication or comorbidity and clinically stable for discharge, ≥ One:**
 - Adverse drug reaction resolving
 - Blood glucose within acceptable limits ⁽¹⁵⁾
 - *C. difficile* infection, symptoms resolving, and tolerating PO ⁽¹⁶⁾

– Tolerating PO ⁽¹⁷⁾

● **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**

- **Use Extended Stay subset**
 - Refer for secondary review
-

Discharge, Level of Care, One: ⁽³⁶⁾

● **HOME, Both:**

● Level of care appropriateness, **Both:**

- Home environment safe and accessible ⁽³⁷⁾
- Patient or caregiver demonstrates ability to manage care

● Discharge planning, **All:**

- Provide comprehensive written discharge and teaching instructions ⁽³⁸⁾
- Medication reconciliation ⁽³⁹⁾
- Follow-up care planned ⁽⁴⁰⁾
- Identify and address transportation needs

● **HOME CARE, Both:**

● Level of care appropriateness, **All:**

- Home environment safe and accessible ⁽³⁷⁾
- Patient or caregiver willing and able to learn care
- Outpatient management contraindicated or unavailable ^(41, 42)

● Skilled services, ≥ **One:** ⁽⁴³⁾

- Skilled nursing ⁽⁴⁴⁾
- Skilled therapy, PT, OT, or ST
- Behavioral health ⁽⁴⁵⁾

● Discharge planning, **All:**

- Provide comprehensive written discharge and teaching instructions ⁽³⁸⁾
- Medication reconciliation ⁽³⁹⁾
- Follow-up care planned ⁽⁴⁰⁾
- Identify and address transportation needs

● **BEHAVIORAL HEALTH, Both:**

● Level of care appropriateness, **One:**

- Support systems available ⁽⁴⁶⁾

● Skilled treatment, ≥ **One:**

- Clinical assessment ⁽⁴⁷⁾
- Individual, group, or family counseling
- Psychiatric, substance, or medication evaluation ⁽⁴⁸⁾

● **SKILLED MEDICAL OR THERAPY, Both:**

● Level of care appropriateness, **All:**

- Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
- Treatment precluded in a lower level

● Services, ≥ **One:**

- Medical and skilled nursing interventions or assessment 1-2x/24h ⁽⁴⁴⁾

● Therapy, **All:**

- Goal directed therapy with rehabilitation potential ^(49, 50)
- Functional impairments requiring at least supervision ⁽⁵¹⁾
- Able to tolerate 1h/d of skilled therapy ≥ 5d/wk

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽³⁹⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **SUBACUTE MEDICAL OR THERAPY, Both:**

- Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 2 x/wk
 - Treatment precluded in a lower level
- Services, $\geq$ **One:**
 - Medical and skilled nursing interventions or assessment 4h/24h ⁽⁴⁴⁾
- Therapy, **All:**
 - Goal directed therapy with rehabilitation potential ^(49, 50)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁵¹⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5 d/wk and ≥ 1 therapy discipline
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽³⁹⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE REHAB, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 3 x/wk
 - Treatment precluded in a lower level
 - Services, **All:**
 - Goal directed therapy with rehabilitation potential ^(49, 50)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁵¹⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5 d/wk and ≥ 2 therapy disciplines
 - Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽³⁹⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **ACUTE REHAB, Both:**
 - Level of care appropriateness, **Both:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 3 x/wk
 - Services, **All:**
 - Goal directed therapy with rehabilitation potential ^(49, 50)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁵¹⁾
 - Able to tolerate ≥ 15 h/wk of skilled therapy and ≥ 2 therapy disciplines ⁽⁵²⁾
 - Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽³⁹⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**
 - Level of care appropriateness, **Both:**
 - Treatment precluded in a lower level
 - In lieu of acute or continued hospitalization or failed lower level of care, $\geq$ **One:**
 - Primary condition or illness and treatment of active comorbid condition(s) ⁽⁵³⁾
 - Ventilator dependent ≥ 6 h/d and weaning planned ^(54, 55)
 - Acute and comorbid conditions requiring prolonged hospitalization ⁽⁵⁶⁾
 - Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽³⁹⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
539 OSTEOMYELITIS WITH MCC	n/a	6.4	CMS GMLOS
540 OSTEOMYELITIS WITH CC	n/a	4.4	CMS GMLOS
541 OSTEOMYELITIS WITHOUT CC/MCC	n/a	3	CMS GMLOS
548 SEPTIC ARTHRITIS WITH MCC	n/a	6.1	CMS GMLOS
549 SEPTIC ARTHRITIS WITH CC	n/a	4.1	CMS GMLOS
550 SEPTIC ARTHRITIS WITHOUT CC/MCC	n/a	3.1	CMS GMLOS
551 MEDICAL BACK PROBLEMS WITH MCC	n/a	4.6	CMS GMLOS
552 MEDICAL BACK PROBLEMS WITHOUT MCC	n/a	2.9	CMS GMLOS
OSTEOMYELITIS	13%	7.5	InterQual
SEPTIC ARTHRITIS	11%	6.1	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:**1:**

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and intensity of service requirements.

For further clarification, see the Acute Level of Care Review Process.

2:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

3:

Erythrocyte sedimentation rate (ESR) and C-reactive protein (CRP) are inflammatory markers. While not specific to the site of inflammation or infection, when elevated, these markers indicate the presence of inflammation. Following ESR and CRP levels is useful when monitoring response to treatment (Paakkonen et al., Clinical orthopaedics and related research 2010, 468: 861-6; Reitzenstein et al., Pediatr Rep 2010, 2: e10).

4:

The diagnosis of septic arthritis would also include septic joints.

5:

Septic arthritis is suspected in patients with joint pain, tenderness, decreased range of motion, and elevated white blood cells in synovial fluid (Margaretten et al., JAMA 2007, 297: 1478-88).

6:

Complete blood count, renal function, erythrocyte sedimentation rate, and C-reactive protein levels should be drawn in patients with infection in order to monitor disease regression as well as drug toxicity.

7:

Anti-infective treatment in a patient with septic arthritis should continue for 4-6 weeks. Transition from parenteral to PO anti-infective treatment may occur after two weeks as long as clinical signs, symptoms, and inflammatory markers are decreasing (Mathews et al., Lancet 2010, 375: 846-55).

8:

In addition to anti-infective therapy, successful management of septic arthritis includes the removal of intra-articular pus. Options for drainage include needle aspiration or surgical aspiration using arthroscopy (Mathews et al., Lancet 2010, 375: 846-55).

9:

Deep vein thrombosis prophylaxis may include the following:

- Anticoagulant therapy - low molecular weight heparin or unfractionated heparin
- Intermittent pneumatic compression (IPC) devices
- Graduated compression stockings

Pharmacologic prophylaxis is preferred over IPC devices (Kearon et al., Chest 2016, 149: 315-52; Hemphill et al., Stroke 2015, 46: 2032-60; Kahn et al., Chest 2012, 141: e195S-226S; Qaseem et al., Ann Intern Med 2011, 155: 625-32; Lim et al., Thorax 2009; 64 Suppl 3: iii1-55; Yancy et al., Circulation 2013, 128(16):e240-319; Jauch et al., Stroke 2013, 44: 870-947; Surawicz et al., Am J Gastroenterol 2013, 108: 478-98; quiz 99; Sachdeva et al., Cochrane Database Syst Rev 2010: CD001484; Sajid et al., Cochrane Database Syst Rev 2012, 5: CD007162; Lindenfeld et al., J Card Fail 2010, 16: e1-194).

Instruction: This criteria point may be applied if the patient is therapeutically anticoagulated.

10:

Prosthetic joint infections are classified based on the timing of the infection. Early-onset infection occurs within three months of joint replacement, delayed-onset occurs within 3-24 months, and late-onset occurs after 24 months. Surgical management of a prosthetic joint infection differs depending on the type of infection. Treatment options for prosthetic joint infections include implant retention or removal, along with appropriate anti-infective therapy (Shuman and Malani, Drugs Aging 2011, 28: 13-26).

11:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

12:

Resolution of infection is indicated by a decrease in temperature, a normalizing of white blood cells and other lab values, and improving imaging studies. If there is a wound present, signs of infection should be resolving (e.g., decrease in wound drainage, less pain at site).

13:

Pain is considered to be adequately controlled when a stable medication regimen and/or adjunct therapy provides sufficient, symptomatic relief without causing significant adverse side effects. Studies have shown epidural analgesia to be very effective in the initial post-operative period (McNicol et al., Cochrane Database Syst Rev 2015: CD003348; Popping et al., Ann Surg 2014, 259: 1056-67). For acute pain, the Center for Disease Control (CDC) recommends prescribing the lowest effective dose of immediate-release opioids, with careful attention paid to the quantity and duration of the prescription. Three days or less will often work; more than 7 days is rare. For chronic pain unrelated to active cancer, palliative or end-of-life care, the CDC recommends the following guidelines for clinicians (Dowell et al., Jama 2016, 315: 1624-45; Dowell et al., MMWR Recomm Rep 2016, 65: 1-49):

- Consider nonpharmacologic therapy and nonopioid pharmacologic therapy first
- Know the patient's history of controlled substance prescriptions, using the state prescription drug monitoring program (DMP) data when initiating opioid therapy and then every 3 months thereafter on every prescription
- Weigh the risks and benefits of opioid therapy before starting or continuing opioids
- Establish realistic treatment goals, discuss risks and benefits of opioid therapy, and go over the plan for discontinuing opioids if the harm outweighs the benefits
- Discuss patient and clinician responsibilities for managing this therapy. If a patient has an opioid use disorder, offer evidence-based, medication-assisted treatment in combination with behavioral therapies

When prescribing opioids for chronic pain:

- Combine them with nonpharmacologic therapy/nonopioid therapy
- Prescribe immediate-release opioids versus extended-release or long-acting opioids
- Start at the lowest dose
- Evaluate the benefits and the harm with patients within 1 to 4 weeks of starting opioid therapy, and then at least every 3 months thereafter
- Consider annual urine testing to assess for prescribed medications as well as other controlled prescription and/or illicit drugs
- Avoid prescribing opioid pain medication and benzodiazepines concurrently

For patients in the early post-surgical phase, a large retrospective cohort study found that the duration of the opioid prescription, rather than its dosage, was more strongly associated with opioid misuse (Brat et al., BMJ 2018, 360: j5790).

14:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

15:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

16:

Signs indicating recovery from *Clostridioides difficile* include:

- Decrease in the severity and frequency of diarrhea
- Absence of fever
- Normal white blood cell count

Not all patients demonstrate fever or leukocytosis during the course of illness. Diet advancement should be based on improving clinical signs and symptoms.

17:

Oral intake should be at least equivalent to the patient's maintenance fluid requirements or within parameters which the medical practitioner deems appropriate. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

18:

The Centers for Disease Control and Prevention (CDC) define a fever as greater than or equal to 100.4 degrees Fahrenheit (i.e., greater than or equal to 38 degrees Celsius) and do not specify the route of measurement (Centers for Disease Control and Prevention, Definitions of symptoms for reportable illness. 2017). The route utilized may be directed by institutional protocols, equipment availability, patient preference, or other patient-specific factors.

19:

For a definitive diagnosis of osteomyelitis a pathogenic organism must be isolated from a bone lesion or blood culture. Blood cultures may be obtained in patients with suspected osteomyelitis and when positive, often eliminate the need for bone culture or biopsy. In patients with chronic osteomyelitis bone cultures are recommended (Calhoun and Manring, Infect Dis Clin North Am 2005, 19: 765-86).

20:

Although the vast majority of wound care is readily handled in the post-acute setting, there are situations when a patient requires care at the acute level, such as high-frequency wound care required, prolonged length of time required performing the procedure, and the type or amount of medication required to keep the patient comfortable during the procedure. For children less than 12 years of age (or less than 30 kg), the wound care regimen may need to be adjusted to a lesser frequency and/or duration (e.g., twice per day, 10 minutes), though still requiring the current level of care, due to the patient's size, pain tolerance, fluid shifts, and vital sign lability.

To determine the appropriate discharge destination, an assessment is required of the patient and/or caregiver's ability to learn the necessary wound care regimen, complexity of the treatment regimen, and a plan for managing contributory factors. For discharge to the home setting, the patient and/or caregiver must be able and willing to participate in the care plan, or there is sufficient agency assistance available to adequately meet the patient's needs. Patient and/or caregiver instruction should include all the following (Bryant and Nix, Acute & chronic wounds: current management concepts, 5th ed. 2015, xiv, 655 p.):

- Wound care regimen that includes a plan to monitor patient-specific contributory factors
- Adjunctive therapies (medications, nutritional plan, and supplements)

- Re-ordering medications and wound management supplies
- When to seek medical care (e.g., signs, symptoms, and complications to report to the health care provider)
- Continued follow-up and support with health care providers

21:

Wound care teaching should include instruction on the following:

- Adjunctive therapies (medications, nutritional plan, and supplements)
- Follow-up plan of care
- Medications and wound management supplies
- When to seek medical care (e.g., signs, symptoms, and complications to report to the health care provider)
- Wound care regimen (including patient-specific factors)

22:

Hyperbaric oxygen (HBO) therapy is the application of one hundred percent oxygen at pressures greater than one atmosphere. The resulting hyperoxic environment promotes antimicrobial effects that may enhance or modulate standard therapy, including:

- Enhancement of leukocyte-killing activity
- Bacterial growth suppression in hyperoxic tissues
- Enhancement of antibiotic effects
- Improvement in tissue repair
- Effects on anaerobic bacteria

Evidence supports the use of HBO in patients with chronic refractory osteomyelitis. Oxygen levels in infected bone are decreased. Breathing one hundred percent oxygen in hyperbaric conditions can normalize or increase tissue oxygenation levels. Increasing oxygen levels in infected bone tissue has also been shown to have a direct suppressive effect on anaerobic organisms. Patients with chronic, refractory osteomyelitis should be treated with HBO for 90-120 minutes daily. After three days of treatment, patients may be managed in the outpatient setting (Murad et al., J Clin Epidemiol 2014, 67: 65-72; Kaide and Khandelwal, Emerg Med Clin North Am 2008, 26: 571-95, xi).

23:

Osteomyelitis is deemed refractory when it is chronic and persists or recurs after appropriate treatment.

24:

Pain assessment includes quality, intensity, location, frequency, duration, and exacerbating or alleviating factors. Uncontrolled pain should be monitored, assessed, and documented. Appropriate medication adjustment and other interventions should be taken to reduce pain.

25:

Instruction: For these criteria, medication adjustment refers to a change in dose, frequency of administration, or change in medication.

26:

For acute pain, the Center for Disease Control (CDC) recommends prescribing the lowest effective dose of immediate-release opioids, with careful attention paid to the quantity and duration of the prescription. Three days or less will often work; more than 7 days is rare. For chronic pain unrelated to active cancer, palliative or end-of-life care, the CDC recommends the following guidelines for clinicians (Dowell et al., MMWR Recomm Rep 2016, 65: 1-49):

- Consider nonpharmacologic therapy and nonopioid pharmacologic therapy first
- Know the patient's history of controlled substance prescriptions, using the state prescription drug monitoring program (DMP) data when initiating opioid therapy and then every 3 months thereafter on every prescription
- Weigh the risks and benefits of opioid therapy before starting or continuing opioids

- Establish realistic treatment goals, discuss risks and benefits of opioid therapy, and go over the plan for discontinuing opioids if the harm outweighs the benefits
- Discuss patient and clinician responsibilities for managing this therapy. If a patient has an opioid use disorder, offer evidence-based medication-assisted treatment in combination with behavioral therapies

When prescribing opioids for chronic pain:

- Combine them with nonpharmacologic therapy/nonopioid therapy
- Prescribe immediate-release opioids versus extended release or long-acting opioids
- Start at the lowest dose
- Evaluate the benefits and the harm with patients within 1 to 4 weeks of starting opioid therapy, and then at least every 3 months thereafter
- Consider annual urine testing to assess for prescribed medications as well as other controlled prescription and/or illicit drugs
- Avoid prescribing opioid pain medication and benzodiazepines concurrently

27:

Anti-infective medications may cause adverse drug reactions. Symptoms of an adverse drug reaction include urticaria, angioedema, bronchospasm, pruritus, rash, vomiting, diarrhea, anaphylaxis, and thrombocytopenia.

28:

Guidelines recommend preprandial glucose targets of 80-130 mg/dL (4.4-7.2 mmol/L) and a random blood glucose target range of 100-180 mg/dL (5.6-10.0 mmol/L) in hospitalized patients. In stable patients with previous tight glycemic control, more stringent targets may be appropriate. In patients with severe comorbidities, a history of hypoglycemia, terminal illness, late onset diabetes, and in older patients, less stringent targets may be appropriate. The preferred method for achieving and maintaining glucose control in non-critically ill patients is scheduled subcutaneous insulin administration. Guidelines discourage the sole use of sliding scale insulin in the inpatient hospital setting. Rather, an insulin regimen consisting of basal, prandial, and correctional insulin is recommended in hospitalized patients (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

29:

Instruction: Insulin adjustments should be based on blood glucose values obtained by lab draw or glucose monitor. Intermittent insulin must be administered at least 3 times in 24 hours. If the patient is on a continuous insulin infusion, the rate must be adjusted at least twice in 24 hours. A dose that is held based on a normal or low blood glucose value should be included in the count of doses administered within the 24-hour period.

30:

Clostridioides difficile is the most common cause of nosocomial infectious diarrhea, particularly in patients receiving anti-infectives (Johnson et al., Clin Infect Dis 2021, 73: e1029-e44).

31:

Clostridioides difficile is the most common cause of nosocomial infectious diarrhea, particularly in patients receiving anti-infectives. Guidelines describe multiple acceptable regimens but do indicate a preference for oral fidaxomicin for patients with mild to moderate disease. Alternative regimens are driven by disease severity and the number of episodes the patient has experienced. Treatment guidelines recommend discontinuation of the inciting anti-infective as soon as possible to prevent recurrence. Other interventions include IV fluids, electrolytes, or oral fluids to maintain hydration. Use of opiates and antidiarrheal medications should be avoided because decreased intestinal motility will exacerbate the infection (Johnson et al., Clin Infect Dis 2021, 73: e1029-e44).

32:

Instruction: This criteria point is intended for anti-infectives initiated for the treatment of *C. difficile* infections. The preferred treatment of *Clostridioides difficile* infections is based on the episode of illness and the severity of the symptoms. The Infectious Diseases Society of America (IDSA) recommends a variety of treatment regimens which are dependent on clinical presentation, including disease severity and how many episodes of disease the patient has experienced (Johnson et al., Clin Infect Dis 2021, 73: e1029-e44).

33:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirements. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit (Scales, Br J Nurs 2014, 23).

34:

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35:

Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

36:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

37:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

38:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

39:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer

orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

40:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

41:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

42:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

43:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

44:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring,

and/or other interventions not listed.

45:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

46:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

47:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

48:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

49:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

50:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

51:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

52:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

53:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

54:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

55:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

56:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.